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(54) **CHANNEL STATE INFORMATION
REFERENCE SIGNAL RESOURCE PAIRS**

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H04L 5/00 (2006.01)

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(2013.01)

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H04L 5/0023; H04L 5/0048; H04L
5/0094; H04B 7/0456

See application file for complete search history.

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Primary Examiner — Chuong A Ngo

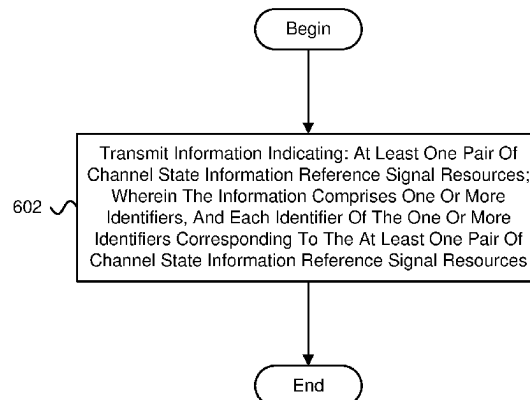
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Adamson

(57) **ABSTRACT**

Apparatuses, methods, and systems are disclosed for chan-
nel state information reference signal resource pairs. One
method (600) includes transmitting (602) information indi-
cating: at least one pair of channel state information refer-
ence signal resources; wherein the information comprises
one or more identifiers, and each identifier of the one or
more identifiers corresponding to the at least one pair of
channel state information reference signal resources.

20 Claims, 9 Drawing Sheets

600
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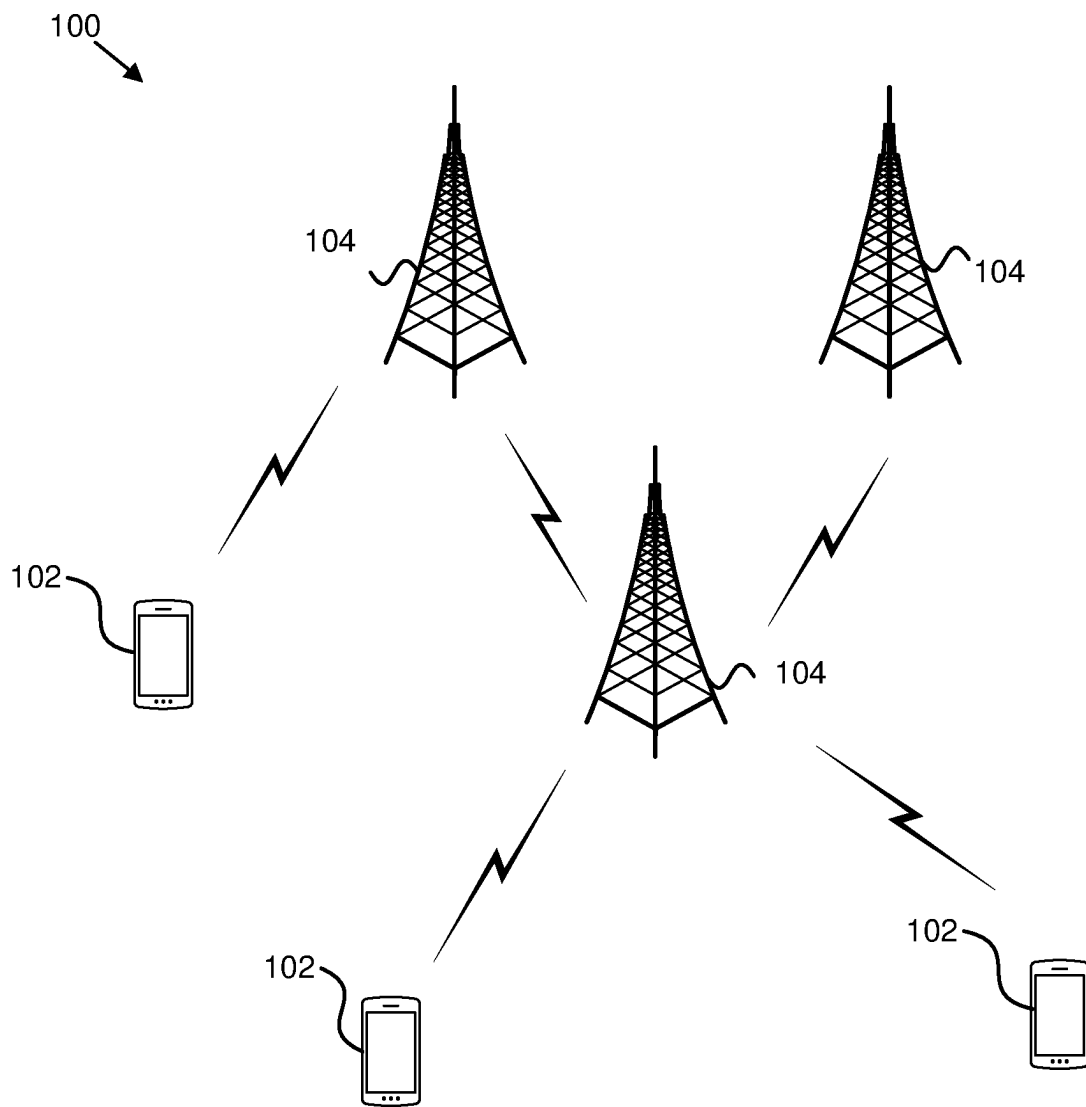


FIG. 1

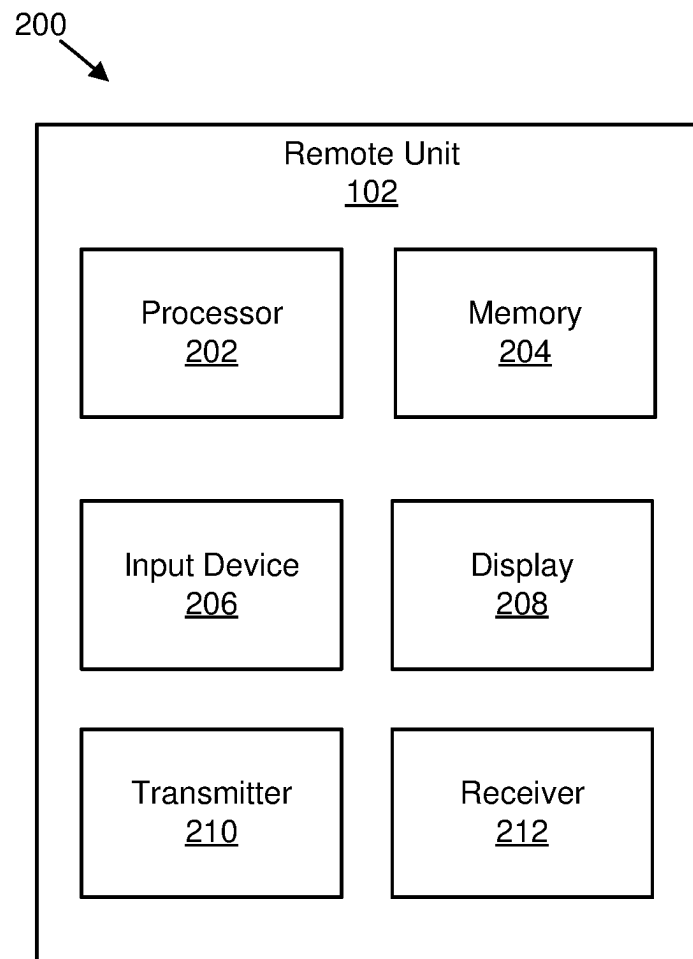


FIG. 2

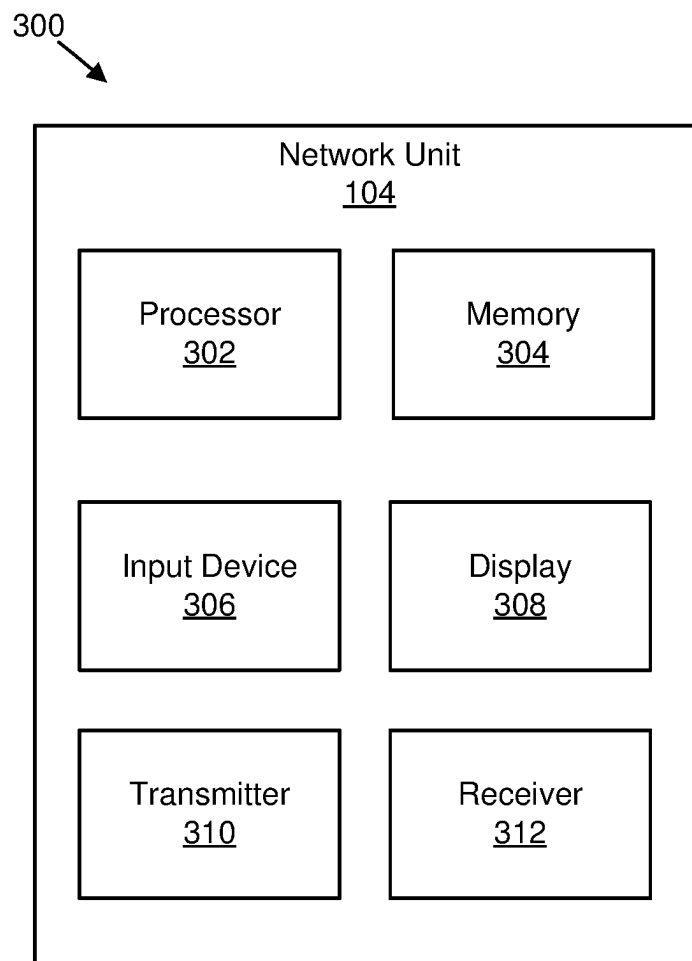


FIG. 3

400

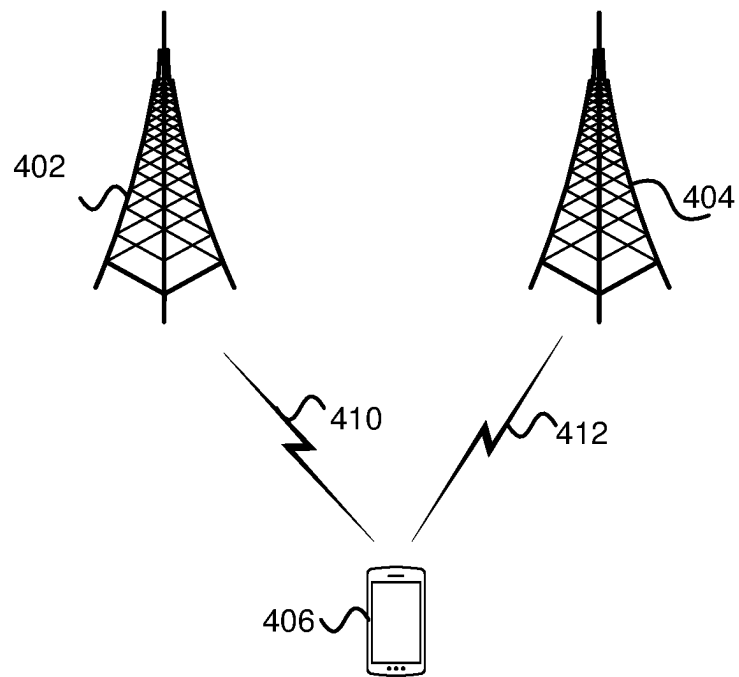



FIG. 4

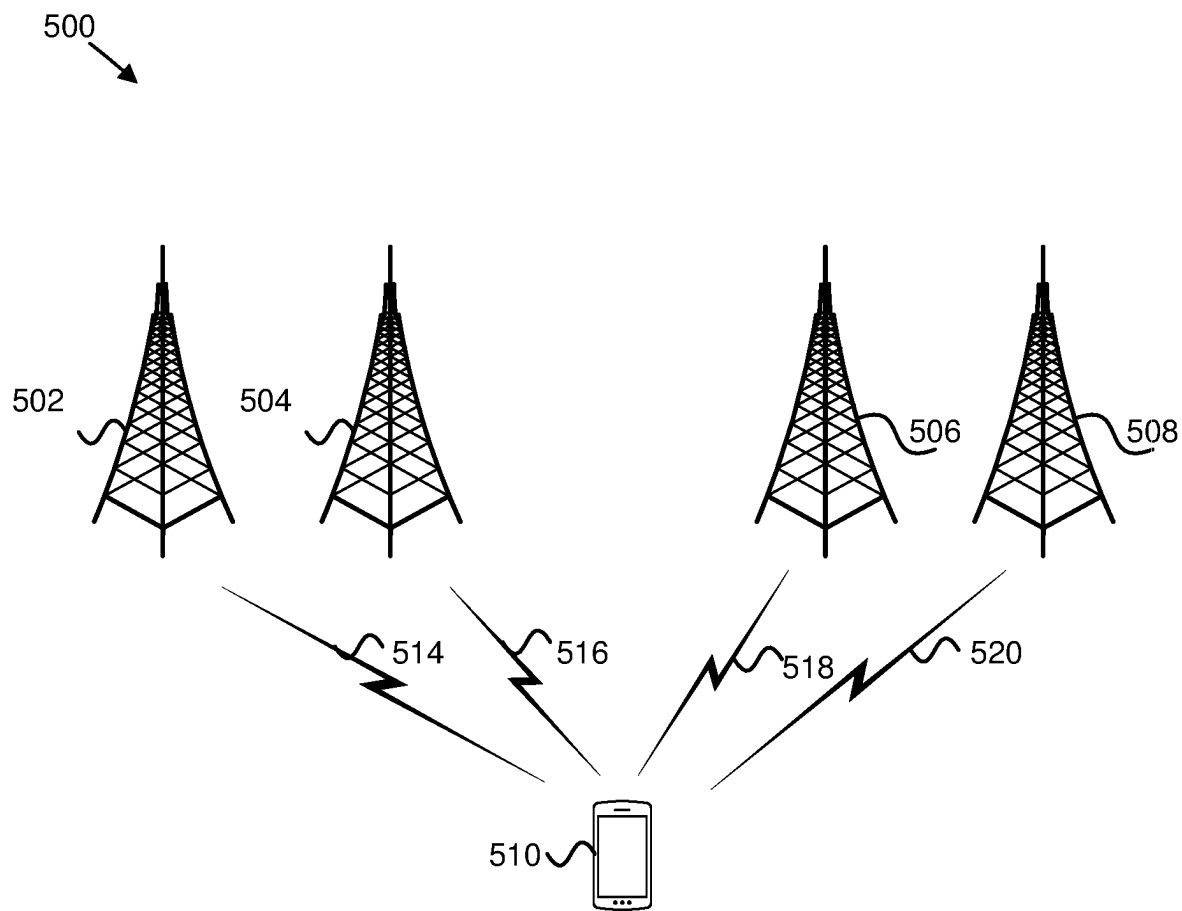


FIG. 5

600 →

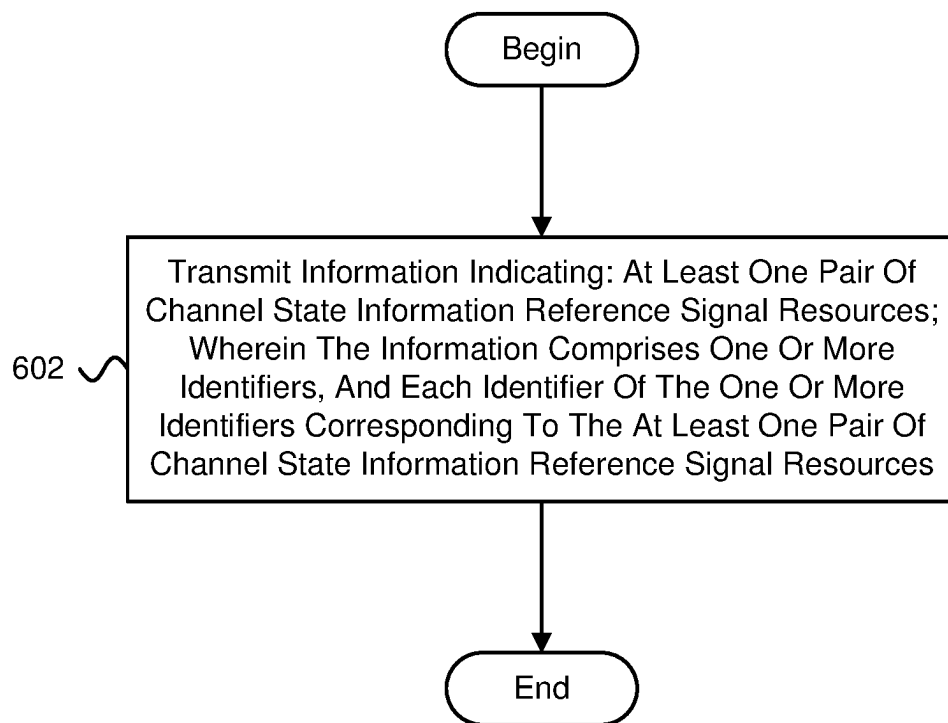


FIG. 6

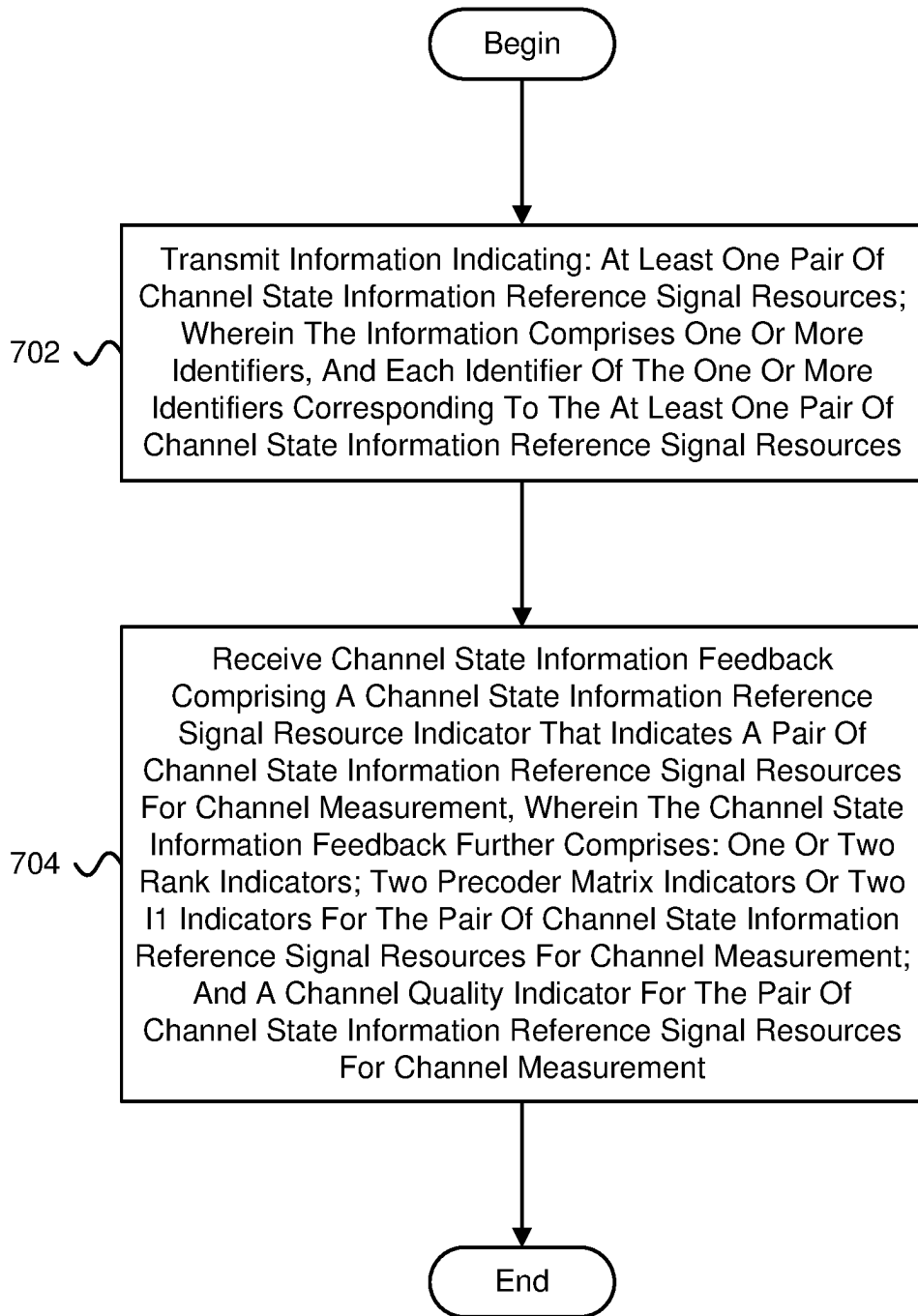
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↓

FIG. 7

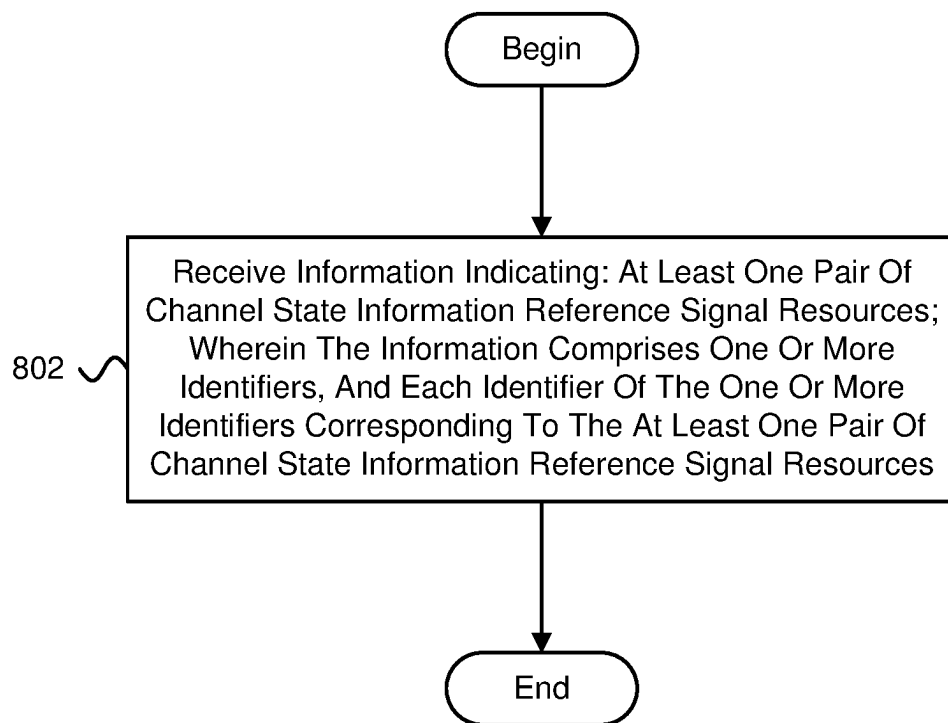
800
↓

FIG. 8

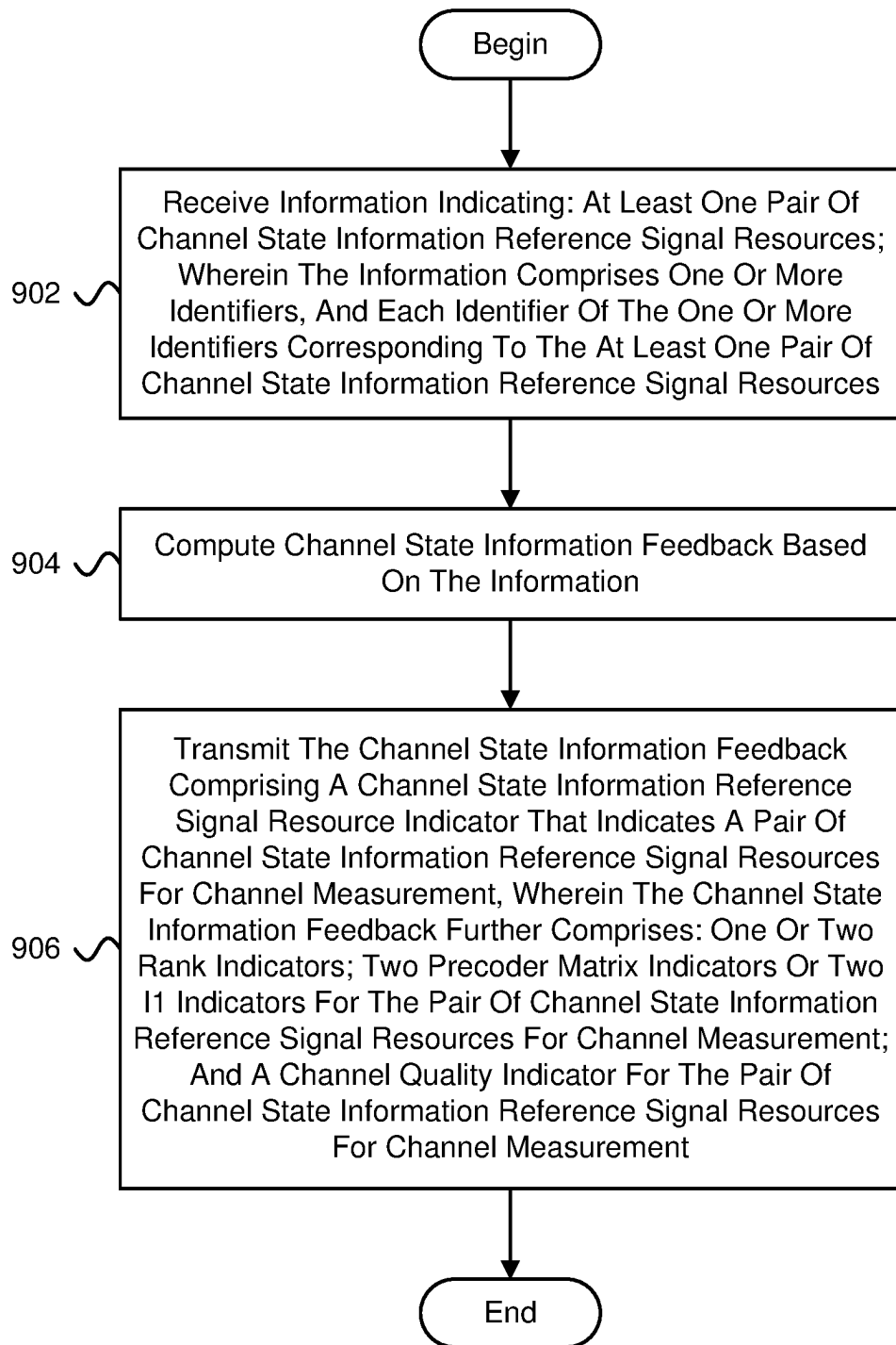
900
↓

FIG. 9

CHANNEL STATE INFORMATION REFERENCE SIGNAL RESOURCE PAIRS

FIELD

The subject matter disclosed herein relates generally to wireless communications and more particularly relates to channel state information reference signal resource pairs.

BACKGROUND

The following abbreviations are herewith defined, at least some of which are referred to within the following description: Third Generation Partnership Project ("3GPP"), 5G QOS Indicator ("5QI"), Acknowledge Mode ("AM"), Aperiodic ("AP"), Backhaul ("BH"), Broadcast Multicast ("BM"), Buffer Occupancy ("BO"), Base Station ("BS"), Buffer Status Report ("BSR"), Bandwidth ("BW"), Bandwidth Part ("BWP"), Carrier Aggregation ("CA"), Component Carrier ("CC"), Code Division Multiplexing ("CDM"), Control Element ("CE"), Coordinated Multipoint ("CoMP"), Categories of Requirements ("CoR"), Control Resource Set ("CORESET"), Cyclic Prefix ("CP"), Cyclic Prefix OFDM ("CP-OFDM"), CSI-RS Resource Indicator ("CRI"), Cell RNTI ("C-RNTI"), Channel State Information ("CSI"), CSI IM ("CSI-IM"), CSI RS ("CSI-RS"), Channel Quality Indicator ("CQI"), Central Unit ("CU"), Codeword ("CW"), Downlink Assignment Index ("DAI"), Downlink Control Information ("DCI"), Downlink ("DL"), Discrete Fourier Transform Spread OFDM ("DFT-s-OFDM"), Demodulation Reference Signal ("DMRS" or "DM-RS"), Data Radio Bearer ("DRB"), Dedicated Short-Range Communications ("DSRC"), Distributed Unit ("DU"), Enhanced Mobile Broadband ("eMBB"), Evolved Node B ("eNB"), Enhanced Subscriber Identification Module ("eSIM"), Enhanced ("E"), Frequency Division Duplex ("FDD"), Frequency Division Multiple Access ("FDMA"), Frequency Range ("FR"), 450 MHz-6000 MHz ("FR1"), 24250 MHz-52600 MHz ("FR2"), Hybrid Automatic Repeat Request ("HARQ"), High-Definition Multimedia Interface ("HDMI"), Integrated Access Backhaul ("IAB"), Identity or Identifier or Identification ("ID"), Information Element ("IE"), Interference Measurement ("IM"), International Mobile Subscriber Identity ("IMSI"), Internet-of-Things ("IoT"), Internet Protocol ("IP"), Joint Transmission ("JT"), Level 1 ("L1"), L1 RSRP ("L1-RSRP"), L1 SINR ("L1-SINR"), Logical Channel ("LCH"), Logical Channel Group ("LCG"), Logical Channel ID ("LCID"), Logical Channel Prioritization ("LCP"), Layer Indicator ("LI"), Long Term Evolution ("LTE"), Levels of Automation ("LoA"), Medium Access Control ("MAC"), Modulation Coding Scheme ("MCS"), Multi DCI ("M-DCI"), Master Information Block ("MIB"), Multiple Input Multiple Output ("MIMO"), Mobile-Termination ("MT"), Machine Type Communication ("MTC"), Multi PDSCH ("Multi-PDSCH"), Multi TRP ("M-TRP"), Multi-User ("MU"), Multi-User MIMO ("MU-MIMO"), Minimum Mean Square Error ("MMSE"), Negative-Acknowledgment ("NACK") or ("NAK"), Next Generation ("NG"), Next Generation Node B ("gNB"), New Radio ("NR"), Non-Zero Power ("NZP"), NZP CSI-RS ("NZP-CSI-RS"), Orthogonal Frequency Division Multiplexing ("OFDM"), Peak-to-Average Power Ratio ("PAPR"), Physical Broadcast Channel ("PBCH"), Physical Downlink Control Channel ("PDCCH"), Physical Downlink Shared Channel ("PDSCH"), PDSCH Configuration ("PDSCH-Config"), Policy Control Function ("PCF"), Packet Data Convergence Protocol ("PDCP"), Packet Data

Network ("PDN"), Protocol Data Unit ("PDU"), Public Land Mobile Network ("PLMN"), Precoding Matrix Indicator ("PMI"), ProSe Per Packet Priority ("PPPP"), ProSe Per Packet Reliability ("PPPR"), Physical Resource Block ("PRB"), Packet Switched ("PS"), Physical Sidelink Control Channel ("PSCCH"), Physical Sidelink Shared Channel ("PSSCH"), Phase Tracking RS ("PTRS" or "PT-RS"), Physical Uplink Control Channel ("PUCCH"), Physical Uplink Shared Channel ("PUSCH"), Quasi Co-Located ("QCL"), Quality of Service ("QoS"), Random Access Channel ("RACH"), Radio Access Network ("RAN"), Radio Access Technology ("RAT"), Resource Element ("RE"), Radio Frequency ("RF"), Rank Indicator ("RI"), Radio Link Control ("RLC"), Radio Link Failure ("RLF"), Radio Network Temporary Identifier ("RNTI"), Resource Pool ("RP"), Radio Resource Control ("RRC"), Remote Radio Head ("RRH"), Reference Signal ("RS"), Reference Signal Received Power ("RSRP"), Reference Signal Received Quality ("RSRQ"), Receive ("RX"), Single Carrier Frequency Domain Spread Spectrum ("SC-FDSS"), Secondary Cell ("SCell"), Sub Carrier Spacing ("SCS"), Single DCI ("S-DCI"), Spatial Division Multiplexing ("SDM"), Service Data Unit ("SDU"), Single Frequency Network ("SFN"), Subscriber Identity Module ("SIM"), Signal-to-Interference Ratio ("SINR"), Sidelink ("SL"), Sequence Number ("SN"), Semi Persistent ("SP"), Scheduling Request ("SR"), SRS Resource Indicator ("SRI"), Sounding Reference Signal ("SRS"), Synchronization Signal ("SS"), SS/PBCH Block ("SSB"), Transport Block ("TB"), Transmission Configuration Indication ("TCI"), Time Division Duplex ("TDD"), Temporary Mobile Subscriber Identity ("TMSI"), Transmitted Precoding Matrix Indicator ("TPMI"), Transmission Reception Point ("TRP"), Technical Standard ("TS"), Transmit ("TX"), User Entity/Equipment (Mobile Terminal) ("UE"), Universal Integrated Circuit Card ("UICC"), Uplink ("UL"), Unacknowledged Mode ("UM"), Universal Mobile Telecommunications System ("UMTS"), LTE Radio Interface ("Uu interface"), User Plane ("UP"), Universal Subscriber Identity Module ("USIM"), Universal Terrestrial Radio Access Network ("UTRAN"), Vehicle to Everything ("V2X"), Voice Over IP ("VOIP"), Visited Public Land Mobile Network ("VPLMN"), Vehicle RNTI ("V-RNTI"), Worldwide Interoperability for Microwave Access ("WiMAX"), Zero Forcing ("ZF"), Zero Power ("ZP"), and ZP CSI-RS ("ZP-CSI-RS"). As used herein, "HARQ-ACK" may represent collectively the Positive Acknowledge ("ACK") and the Negative Acknowledge ("NAK"). ACK means that a TB is correctly received while NAK means a TB is erroneously received.

In certain wireless communications networks, UEs may travel at high speeds.

BRIEF SUMMARY

Methods for channel state information reference signal resource pairs are disclosed. Apparatuses and systems also perform the functions of the methods. In one embodiment, the method includes transmitting information indicating: at least one pair of channel state information reference signal resources; wherein the information comprises one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources.

An apparatus for channel state information reference signal resource pairs, in one embodiment, includes a transmitter that transmits information indicating: at least one pair

of channel state information reference signal resources; wherein the information comprises one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources.

A method for channel state information reference signal resource pairs includes receiving information indicating: at least one pair of channel state information reference signal resources; wherein the information comprises one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources.

An apparatus for channel state information reference signal resource pairs, in one embodiment, includes a receiver that receives information indicating: at least one pair of channel state information reference signal resources; wherein the information comprises one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources.

BRIEF DESCRIPTION OF THE DRAWINGS

A more particular description of the embodiments briefly described above will be rendered by reference to specific embodiments that are illustrated in the appended drawings. Understanding that these drawings depict only some embodiments and are not therefore to be considered to be limiting of scope, the embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings, in which:

FIG. 1 is a schematic block diagram illustrating one embodiment of a wireless communication system for channel state information reference signal resource pairs;

FIG. 2 is a schematic block diagram illustrating one embodiment of an apparatus that may be used for channel state information reference signal resource pairs;

FIG. 3 is a schematic block diagram illustrating one embodiment of an apparatus that may be used for channel state information reference signal resource pairs;

FIG. 4 is a schematic block diagram illustrating one embodiment of a system including communications with a UE traveling at high speeds;

FIG. 5 is a schematic block diagram illustrating another embodiment of a system including communications with a UE traveling at high speeds;

FIG. 6 is a schematic flow chart diagram illustrating one embodiment of a method for channel state information reference signal resource pairs;

FIG. 7 is a schematic flow chart diagram illustrating another embodiment of a method for channel state information reference signal resource pairs;

FIG. 8 is a schematic flow chart diagram illustrating a further embodiment of a method for channel state information reference signal resource pairs; and

FIG. 9 is a schematic flow chart diagram illustrating yet another embodiment of a method for channel state information reference signal resource pairs.

DETAILED DESCRIPTION

As will be appreciated by one skilled in the art, aspects of the embodiments may be embodied as a system, apparatus, method, or program product. Accordingly, embodiments may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining

software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.” Furthermore, embodiments may take the form of a program product embodied in one or more computer readable storage devices storing machine readable code, computer readable code, and/or program code, referred to hereafter as code. The storage devices may be tangible, non-transitory, and/or non-transmission. The storage devices may not embody signals. In a certain embodiment, the storage devices only employ signals for accessing code.

Certain of the functional units described in this specification may be labeled as modules, in order to more particularly emphasize their implementation independence. For example, a module may be implemented as a hardware circuit comprising custom very-large-scale integration (“VLSI”) circuits or gate arrays, off-the-shelf semiconductor devices such as logic chips, transistors, or other discrete components. A module may also be implemented in programmable hardware devices such as field programmable gate arrays, programmable array logic, programmable logic devices or the like.

Modules may also be implemented in code and/or software for execution by various types of processors. An identified module of code may, for instance, include one or more physical or logical blocks of executable code which may, for instance, be organized as an object, procedure, or function. Nevertheless, the executables of an identified module need not be physically located together, but may include disparate instructions stored in different locations which, when joined logically together, include the module and achieve the stated purpose for the module.

Indeed, a module of code may be a single instruction, or many instructions, and may even be distributed over several different code segments, among different programs, and across several memory devices. Similarly, operational data may be identified and illustrated herein within modules, and may be embodied in any suitable form and organized within any suitable type of data structure. The operational data may be collected as a single data set, or may be distributed over different locations including over different computer readable storage devices. Where a module or portions of a module are implemented in software, the software portions are stored on one or more computer readable storage devices.

Any combination of one or more computer readable medium may be utilized. The computer readable medium may be a computer readable storage medium. The computer readable storage medium may be a storage device storing the code. The storage device may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, holographic, micromechanical, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing.

More specific examples (a non-exhaustive list) of the storage device would include the following: an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random access memory (“RAM”), a read-only memory (“ROM”), an erasable programmable read-only memory (“EPROM” or Flash memory), a portable compact disc read-only memory (“CD-ROM”), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

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Code for carrying out operations for embodiments may be any number of lines and may be written in any combination of one or more programming languages including an object oriented programming language such as Python, Ruby, Java, Smalltalk, C++, or the like, and conventional procedural programming languages, such as the “C” programming language, or the like, and/or machine languages such as assembly languages. The code may execute entirely on the user’s computer, partly on the user’s computer, as a stand-alone software package, partly on the user’s computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user’s computer through any type of network, including a local area network (“LAN”) or a wide area network (“WAN”), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

Reference throughout this specification to “one embodiment,” “an embodiment,” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, appearances of the phrases “in one embodiment,” “in an embodiment,” and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment, but mean “one or more but not all embodiments” unless expressly specified otherwise. The terms “including,” “comprising,” “having,” and variations thereof mean “including but not limited to,” unless expressly specified otherwise. An enumerated listing of items does not imply that any or all of the items are mutually exclusive, unless expressly specified otherwise. The terms “a,” “an,” and “the” also refer to “one or more” unless expressly specified otherwise.

Furthermore, the described features, structures, or characteristics of the embodiments may be combined in any suitable manner. In the following description, numerous specific details are provided, such as examples of programming, software modules, user selections, network transactions, database queries, database structures, hardware modules, hardware circuits, hardware chips, etc., to provide a thorough understanding of embodiments. One skilled in the relevant art will recognize, however, that embodiments may be practiced without one or more of the specific details, or with other methods, components, materials, and so forth. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring aspects of an embodiment.

Aspects of the embodiments are described below with reference to schematic flowchart diagrams and/or schematic block diagrams of methods, apparatuses, systems, and program products according to embodiments. It will be understood that each block of the schematic flowchart diagrams and/or schematic block diagrams, and combinations of blocks in the schematic flowchart diagrams and/or schematic block diagrams, can be implemented by code. The code may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the schematic flowchart diagrams and/or schematic block diagrams block or blocks.

The code may also be stored in a storage device that can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions stored in the storage device

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produce an article of manufacture including instructions which implement the function/act specified in the schematic flowchart diagrams and/or schematic block diagrams block or blocks.

The code may also be loaded onto a computer, other programmable data processing apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatus or other devices to produce a computer implemented process such that the code which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

The schematic flowchart diagrams and/or schematic block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of apparatuses, systems, methods and program products according to various embodiments. In this regard, each block in the schematic flowchart diagrams and/or schematic block diagrams may represent a module, segment, or portion of code, which includes one or more executable instructions of the code for implementing the specified logical function(s).

It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. Other steps and methods may be conceived that are equivalent in function, logic, or effect to one or more blocks, or portions thereof, of the illustrated Figures.

Although various arrow types and line types may be employed in the flowchart and/or block diagrams, they are understood not to limit the scope of the corresponding embodiments. Indeed, some arrows or other connectors may be used to indicate only the logical flow of the depicted embodiment. For instance, an arrow may indicate a waiting or monitoring period of unspecified duration between enumerated steps of the depicted embodiment. It will also be noted that each block of the block diagrams and/or flowchart diagrams, and combinations of blocks in the block diagrams and/or flowchart diagrams, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and code.

The description of elements in each figure may refer to elements of proceeding figures. Like numbers refer to like elements in all figures, including alternate embodiments of like elements.

FIG. 1 depicts an embodiment of a wireless communication system **100** for channel state information reference signal resource pairs. In one embodiment, the wireless communication system **100** includes remote units **102** and network units **104**. Even though a specific number of remote units **102** and network units **104** are depicted in FIG. 1, one of skill in the art will recognize that any number of remote units **102** and network units **104** may be included in the wireless communication system **100**.

In one embodiment, the remote units **102** may include computing devices, such as desktop computers, laptop computers, personal digital assistants (“PDAs”), tablet computers, smart phones, smart televisions (e.g., televisions connected to the Internet), set-top boxes, game consoles, security systems (including security cameras), vehicle on-board computers, network devices (e.g., routers, switches, modems), IoT devices, or the like. In some embodiments, the remote units **102** include wearable devices, such as smart

watches, fitness bands, optical head-mounted displays, or the like. Moreover, the remote units **102** may be referred to as subscriber units, mobiles, mobile stations, users, terminals, mobile terminals, fixed terminals, subscriber stations, UE, user terminals, a device, or by other terminology used in the art. The remote units **102** may communicate directly with one or more of the network units **104** via UL communication signals and/or the remote units **102** may communicate directly with other remote units **102** via sidelink communication.

The network units **104** may be distributed over a geographic region. In certain embodiments, a network unit **104** may also be referred to as an access point, an access terminal, a base, a base station, a Node-B, an eNB, a gNB, a Home Node-B, a RAN, a relay node, a device, a network device, an IAB node, a donor IAB node, or by any other terminology used in the art. The network units **104** are generally part of a radio access network that includes one or more controllers communicably coupled to one or more corresponding network units **104**. The radio access network is generally communicably coupled to one or more core networks, which may be coupled to other networks, like the Internet and public switched telephone networks, among other networks. These and other elements of radio access and core networks are not illustrated but are well known generally by those having ordinary skill in the art.

In one implementation, the wireless communication system **100** is compliant with the 5G or NG (Next Generation) standard of the 3GPP protocol, wherein the network unit **104** transmits using NG RAN technology. More generally, however, the wireless communication system **100** may implement some other open or proprietary communication protocol, for example, WiMAX, among other protocols. The present disclosure is not intended to be limited to the implementation of any particular wireless communication system architecture or protocol.

The network units **104** may serve a number of remote units **102** within a serving area, for example, a cell or a cell sector via a wireless communication link. The network units **104** transmit DL communication signals to serve the remote units **102** in the time, frequency, and/or spatial domain.

In some embodiments, a network unit **104** may transmit information indicating: at least one pair of channel state information reference signal resources; wherein the information comprises one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources. Accordingly, a network unit **104** may be used for channel state information reference signal resource pairs.

In various embodiments, a remote unit **102** may receive information indicating: at least one pair of channel state information reference signal resources; wherein the information comprises one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources. Accordingly, a remote unit **102** may be used for channel state information reference signal resource pairs.

FIG. 2 depicts one embodiment of an apparatus **200** that may be used for channel state information reference signal resource pairs. The apparatus **200** includes one embodiment of the remote unit **102**. Furthermore, the remote unit **102** may include a processor **202**, a memory **204**, an input device **206**, a display **208**, a transmitter **210**, and a receiver **212**. In some embodiments, the input device **206** and the display **208** are combined into a single device, such as a touchscreen. In certain embodiments, the remote unit **102** may not include any input device **206** and/or display **208**. In various embodi-

ments, the remote unit **102** may include one or more of the processor **202**, the memory **204**, the transmitter **210**, and the receiver **212**, and may not include the input device **206** and/or the display **208**.

The processor **202**, in one embodiment, may include any known controller capable of executing computer-readable instructions and/or capable of performing logical operations. For example, the processor **202** may be a microcontroller, a microprocessor, a central processing unit ("CPU"), a graphics processing unit ("GPU"), an auxiliary processing unit, a field programmable gate array ("FPGA"), or similar programmable controller. In some embodiments, the processor **202** executes instructions stored in the memory **204** to perform the methods and routines described herein. The processor **202** is communicatively coupled to the memory **204**, the input device **206**, the display **208**, the transmitter **210**, and the receiver **212**.

The memory **204**, in one embodiment, is a computer readable storage medium. In some embodiments, the memory **204** includes volatile computer storage media. For example, the memory **204** may include a RAM, including dynamic RAM ("DRAM"), synchronous dynamic RAM ("SDRAM"), and/or static RAM ("SRAM"). In some embodiments, the memory **204** includes non-volatile computer storage media. For example, the memory **204** may include a hard disk drive, a flash memory, or any other suitable non-volatile computer storage device. In some embodiments, the memory **204** includes both volatile and non-volatile computer storage media. In some embodiments, the memory **204** also stores program code and related data, such as an operating system or other controller algorithms operating on the remote unit **102**.

The input device **206**, in one embodiment, may include any known computer input device including a touch panel, a button, a keyboard, a stylus, a microphone, or the like. In some embodiments, the input device **206** may be integrated with the display **208**, for example, as a touchscreen or similar touch-sensitive display. In some embodiments, the input device **206** includes a touchscreen such that text may be input using a virtual keyboard displayed on the touchscreen and/or by handwriting on the touchscreen. In some embodiments, the input device **206** includes two or more different devices, such as a keyboard and a touch panel.

The display **208**, in one embodiment, may include any known electronically controllable display or display device. The display **208** may be designed to output visual, audible, and/or haptic signals. In some embodiments, the display **208** includes an electronic display capable of outputting visual data to a user. For example, the display **208** may include, but is not limited to, an LCD display, an LED display, an OLED display, a projector, or similar display device capable of outputting images, text, or the like to a user. As another, non-limiting, example, the display **208** may include a wearable display such as a smart watch, smart glasses, a heads-up display, or the like. Further, the display **208** may be a component of a smart phone, a personal digital assistant, a television, a table computer, a notebook (laptop) computer, a personal computer, a vehicle dashboard, or the like.

In certain embodiments, the display **208** includes one or more speakers for producing sound. For example, the display **208** may produce an audible alert or notification (e.g., a beep or chime). In some embodiments, the display **208** includes one or more haptic devices for producing vibrations, motion, or other haptic feedback. In some embodiments, all or portions of the display **208** may be integrated with the input device **206**. For example, the input device **206** and display **208** may form a touchscreen or similar touch-

sensitive display. In other embodiments, the display **208** may be located near the input device **206**.

In various embodiments, the receiver **212** may receive information indicating: at least one pair of channel state information reference signal resources; wherein the information comprises one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources. Although only one transmitter **210** and one receiver **212** are illustrated, the remote unit **102** may have any suitable number of transmitters **210** and receivers **212**. The transmitter **210** and the receiver **212** may be any suitable type of transmitters and receivers. In one embodiment, the transmitter **210** and the receiver **212** may be part of a transceiver.

FIG. 3 depicts one embodiment of an apparatus **300** that may be used for channel state information reference signal resource pairs. The apparatus **300** includes one embodiment of the network unit **104**. Furthermore, the network unit **104** may include a processor **302**, a memory **304**, an input device **306**, a display **308**, a transmitter **310**, and a receiver **312**. As may be appreciated, the processor **302**, the memory **304**, the input device **306**, the display **308**, the transmitter **310**, and the receiver **312** may be substantially similar to the processor **202**, the memory **204**, the input device **206**, the display **208**, the transmitter **210**, and the receiver **212** of the remote unit **102**, respectively.

In some embodiments, the transmitter **310** may transmit information indicating: at least one pair of channel state information reference signal resources; wherein the information comprises one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources. Although only one transmitter **310** and one receiver **312** are illustrated, the network unit **104** may have any suitable number of transmitters **310** and receivers **312**. The transmitter **310** and the receiver **312** may be any suitable type of transmitters and receivers. In one embodiment, the transmitter **310** and the receiver **312** may be part of a transceiver.

In certain embodiments, systems with high speed trains may have a single frequency network deployment.

FIG. 4 is a schematic block diagram illustrating one embodiment of a system **400** including communications with a UE traveling at high speeds. The system **400** (e.g., a cell) includes a first RRH **402** (e.g., TRP) and a second RRH **404** (e.g., TRP) that communicate with a UE **406**. The first RRH **402** and the second RRH **404** are deployed along a path (e.g., railway line) on which the high speed vehicle **408** travels. In some embodiments, the first RRH **402** and the second RRH **404** share the same cell ID. In various embodiments, the first RRH **402** and the second RRH **404** are connected to a gNB with an ideal backhaul and may jointly transmit DL data to the UE **406** in PDSCH. Specifically, the first RRH **402** uses a first communication channel **410** (e.g., one or more communication channels) and the second RRH **404** uses a second communication channel **412** (e.g., one or more communication channels).

FIG. 5 is a schematic block diagram illustrating another embodiment of a system **500** including communications with a UE traveling at high speeds. The system **500** (e.g., a cell) includes a first RRH **502** (e.g., TRP), a second RRH **504** (e.g., TRP), a third RRH **506** (e.g., TRP), and a fourth RRH **508** (e.g., TRP) that communicate with a UE **510**. The first RRH **502**, the second RRH **504**, the third RRH **506**, and the fourth RRH **508** are deployed along a path. The first RRH **502** uses a first communication channel **514** (e.g., one

or more communication channels) to transmit CSI-RS (e.g., CSI-RS1) in a first CSI-RS resource, the second RRH **504** uses a second communication channel **516** (e.g., one or more communication channels) to transmit CSI-RS (e.g., CSI-RS2) in a second CSI-RS resource, the third RRH **506** uses a third communication channel **518** (e.g., one or more communication channels) to transmit CSI-RS (e.g., CSI-RS3) in a third CSI-RS resource, and the fourth RRH **508** uses a fourth communication channel **520** (e.g., one or more communication channels) to transmit CSI-RS (e.g., CSI-RS4) in a fourth CSI-RS resource. The different RRHs transmit different CSI-RS in different CSI-RS resources. Different CSI-RS resources are configured for different RRHs. By separating CSI-RS signals from different RRHs in a time domain and/or frequency domain, the UE **510** may distinguish between different DL signals from different RRHs.

In some embodiments, transmission of PDSCH may be done through pairs of adjacent RRHs. For example, for an RRH pair including the first RRH **502** and the second RRH **504**, the first RRH **502** is configured to transmit CSI-RS in a first CSI-RS resource and the second RRH **504** is configured to transmit CSI-RS in a second CSI-RS resource. PDSCH may be transmitted from two adjacent RRHs to the UE **510** using various transmissions schemes, such as using a first transmission scheme (e.g., SFN from two RRHs in all layers of data (e.g., transmission rank)) described in relation to a first embodiment, and a second transmission scheme (e.g., SDM transmissions from two RRHs) described in relation to a second embodiment.

In the first embodiment, all layers in PDSCH are transmitted jointly from both RRHs in a pair of RRHs. This may enable a strength of each layer to not vary significantly as the UE **510** moves from between the two RRHs (e.g., moving from the first RRH **502** (back RRH) toward the second RRH **504** (front RRH)). In the first embodiment, a DMRS of each layer is associated with two CSI-RS resources for a QCL assumption (e.g., the two CSI-RS resources are QCL). In certain configurations of the first embodiment: 1) each PDSCH layer is transmitted by two RRHs (e.g., a pair of adjacent RRHs); 2) PDSCH is transmitted by the two RRHs sharing the same DMRS ports (e.g., the same DMRS port number); 3) each DMRS port of each layer is associated with two CSI-RS resources (DMRS ports are in one or two CDM groups, depending on the number of DMRS ports); and/or 4) a TCI field in DCI format 1_1 indicates two TCI states, each of which is associated with a CSI-RS resource as QCL type A or D (e.g., QCL type A may correspond to {Doppler shift, Doppler spread, average delay, delay spread}—there may be a QCL correspondence between Doppler shift, Doppler spread, average delay, and delay spread, while QCL type D may correspond to {spatial RX parameter}—there may be a QCL correspondence for the spatial RX parameter) and both TCI states are mapped to the same set of DMRS ports.

In the second embodiment, each layer in PDSCH is transmitted from only one RRH. This may enable the UE **510** to separate a signal from different RRHs with different Doppler shift in a spatial domain. In certain configurations of the second embodiment: 1) each PDSCH layer is transmitted by one of two adjacent RRHs, where all the layers transmitted from an RRH (e.g., TRP) are in a same CDM group or groups; 2) DMRS ports transmitted from an RRH are associated with one CSI-RS resource or two CSI-RS resources with different QCL type; and/or 3) a TCI field in DCI format 1_1 indicates two TCI states, each of which is associated with a CSI-RS resource as QCL type A or D.

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In various embodiments, the UE 510 is configured with multiple (e.g., one or more) CSI report configurations. In such embodiments, each CSI report configuration may have two CSI-RS resources (e.g., or pairs of CSI-RS resources) for channel measurement transmitted from a pair of adjacent RRHs. Based on these two CSI-RS resources for channel measurement, the UE 510 may compute CSI feedback as follows: {RI, PMI1, PMI2, CQI} for the first embodiment; or {RI1, RI2, PMI1, PMI2, CQI}, or {RI, PMI1, PMI2, CQI} for the second embodiment. RI1 and PMI1 are for a first configured CSI-RS resource, and RI2 and PMI2 are for a second configured CSI-RS resource. In some embodiments, PMI may only be configured as i1. If only one RI is included in the CSI feedback, it applies to both CSI-RS resources (RI1=RI2=RI). RI1, RI2, PMI1, PMI2, and CQI are derived jointly from the two CSI-RS resources. If the total rank from two RRHs ($RI_{total}=RI$ for the first embodiment, and $RI_{total}=RI1+RI2$ or $2*RI$ for the second embodiment) is larger than 4, two CQIs (e.g., CQI1, CQI2) may be computed, one for each codeword. For the first embodiment and the second embodiment, if computing CSI feedback, the UE 510 may choose RI, PMI, and CQI to maximize a total PDSCH capacity.

In such embodiments: A) one or more pairs of NZP CSI-RS resources may be configured in CSI-ReportConfig for channel measurement for which: {1) a pair of NZP CSI-RS resources for channel measurement may be identified by a first CSI-RS resource ID if they are arranged as an ordered pair (CSI-RS1, CSI-RS2), or may be identified by a separate index (an ID for a CSI-RS resource pair may be used in the place of CRI in CSI feedback); 2) for a zero power CSI-IM resource configured for interference measurement, it may follow a predetermined behavior; 3) if a resource for interference measurement based on NZP CSI-RS is configured and if pairs of NZP CSI-RS are configured, the UE computes the CSI feedback using the pair of NZP CSI-RS for interference measurement corresponding to the pair of CSI-RS for channel measurement as interference (the UE 510 may assume that a pair of NZP-CSI-RS resources for interference measurement are QCLed with a pair of NZP-CSI-RS resources for channel measurement when computing CSI feedback); 4) for each measurement, an association between a resource setting for channel measurement and a resource setting for interference measurement (CSI-IM, NZP CSI-RS) may follow a predetermined behavior}; B) one or more pairs of CSI-RS resources for channel measurement and interference measurement may be introduced in CSI-ResourceConfig; C) joint PMI1 & PMI2 may be computed based on a pair of CSI-RS resources for channel measurement and/or a pair of CSI-RS resources for interference measurement; and/or D) in CSI feedback: {1) both PMI1 and PMI2 may be included in a CSI feedback in place of PMI or i1 (e.g., i1 may refer to a subset of indicators from a larger set of indicators-a first index out of two indexes) (e.g., $cir-RI-i1-1$ and $cri-RI-i1-2$ may be configured to be reported in one CSI-ReportConfig); and/or 2) CQI reflects a channel quality if both PMI1 and PMI2 are used (e.g., one or two CQIs may be included for one or two codewords)}.

In some embodiments, if a gNB receives a CSI report from the UE 510, the gNB may schedule PDSCH transmission scheme according to the first embodiment or the second embodiment based on CSI feedback.

FIG. 6 is a schematic flow chart diagram illustrating one embodiment of a method 600 for channel state information reference signal resource pairs. In some embodiments, the method 600 is performed by an apparatus, such as the

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network unit 104. In certain embodiments, the method 600 may be performed by a processor executing program code, for example, a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or the like.

The method 600 may include transmitting 602 information indicating: at least one one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources.

In certain embodiments, the information comprises a channel state information resource configuration. In some embodiments, the information comprises a channel state information report configuration. In various embodiments, the channel state information report configuration configures one or more pairs of channel state information reference signal resources for channel measurement.

In one embodiment, the channel state information report configuration configures one or more pairs of non-zero power channel state information reference signal resources for interference measurement, one or more channel state information interference measurement resources for interference measurement, or a combination thereof. In certain embodiments, the channel state information report configuration configures a report quantity field comprising one or two rank indicators, two precoder matrix indicators or two i1 indicators, and a channel quality indicator.

In various embodiments, the method 600 further comprises transmitting a transmission mode indicator that indicates a mode for transmitting information to a user equipment using a pair of transmission reception points. In one embodiment, the method 600 further comprises receiving channel state information feedback comprising a channel state information reference signal resource indicator that indicates a pair of channel state information reference signal resources for channel measurement, wherein the channel state information feedback further comprises: one or two rank indicators; two precoder matrix indicators or two i1 indicators for the pair of channel state information reference signal resources for channel measurement; and a channel quality indicator for the pair of channel state information reference signal resources for channel measurement. In certain embodiments, the method 600 further comprises scheduling a downlink transmission based on the channel state information feedback.

FIG. 7 is a schematic flow chart diagram illustrating another embodiment of a method 700 for channel state information reference signal resource pairs. In some embodiments, the method 700 is performed by an apparatus, such as the network unit 104. In certain embodiments, the method 700 may be performed by a processor executing program code, for example, a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or the like.

The method 700 may include transmitting 702 information indicating: at least one one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources. The method 700 also includes receiving 704 channel state information feedback comprising a channel state information reference signal resource indicator that indicates a pair of channel state information reference signal resources for channel measurement, wherein the channel state information feedback further comprises: one or two rank indicators; two precoder matrix indicators or two i1 indicators for the pair of channel state information reference signal resources for channel measurement; and a channel

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quality indicator for the pair of channel state information reference signal resources for channel measurement.

In certain embodiments, the information comprises a channel state information resource configuration. In some embodiments, the information comprises a channel state information report configuration. In various embodiments, the channel state information report configuration configures one or more pairs of channel state information reference signal resources for channel measurement.

In one embodiment, the channel state information report configuration configures one or more pairs of non-zero power channel state information reference signal resources for interference measurement, one or more channel state information interference measurement resources for interference measurement, or a combination thereof. In certain embodiments, the channel state information report configuration configures a report quantity field comprising one or two rank indicators, two precoder matrix indicators or two i1 indicators, and a channel quality indicator.

In various embodiments, the method 700 further comprises transmitting a transmission mode indicator that indicates a mode for transmitting information to a user equipment using a pair of transmission reception points. In certain embodiments, the method 700 further comprises scheduling a downlink transmission based on the channel state information feedback.

FIG. 8 is a schematic flow chart diagram illustrating a further embodiment of a method 800 for channel state information reference signal resource pairs. In some embodiments, the method 800 is performed by an apparatus, such as the remote unit 102. In certain embodiments, the method 800 may be performed by a processor executing program code, for example, a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or the like.

The method 800 may include receiving 802 information indicating: at least one one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources.

In certain embodiments, the information comprises a channel state information resource configuration. In some embodiments, the information comprises a channel state information report configuration. In various embodiments, the channel state information report configuration configures one or more pairs of channel state information reference signal resources for channel measurement.

In one embodiment, the channel state information report configuration configures one or more pairs of non-zero power channel state information reference signal resources for interference measurement, one or more channel state information interference measurement resources for interference measurement, or a combination thereof. In certain embodiments, the channel state information report configuration configures a report quantity field comprising one or two rank indicators, two precoder matrix indicators or two i1 indicators, and a channel quality indicator.

In various embodiments, the method 800 further comprises receiving a transmission mode indicator that indicates a mode for transmitting information to a user equipment using a pair of transmission configuration indication states. In one embodiment, the method 800 further comprises transmitting channel state information feedback comprising a channel state information reference signal resource indicator that indicates a pair of channel state information reference signal resources for channel measurement, wherein the channel state information feedback further

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comprises: one or two rank indicators; two precoder matrix indicators or two i1 indicators for the pair of channel state information reference signal resources for channel measurement; and a channel quality indicator for the pair of channel state information reference signal resources for channel measurement.

In certain embodiments, the one or two rank indicators, the two precoder matrix indicators or the two i1 indicators, and the channel quality indicator are determined based on the pair of channel state information reference signal resources for channel measurement, and a pair of non-zero power channel state information reference signal resources, a channel state information interference measurement resource, or a combination thereof. In some embodiments, the method 800 further comprises applying a quasi-collocation assumption of the pair of channel state information reference signal resources for channel measurement to a pair of non-zero power channel state information reference signal resources for interference measurement.

In various embodiments, a channel state information reference signal resource identifier of the pair of channel state information reference signal resources for channel measurement is used as the channel state information reference signal resource indicator. In one embodiment, the channel state information reference signal resource identifier occurs only once in the information.

FIG. 9 is a schematic flow chart diagram illustrating yet another embodiment of a method 900 for channel state information reference signal resource pairs. In some embodiments, the method 900 is performed by an apparatus, such as the remote unit 102. In certain embodiments, the method 900 may be performed by a processor executing program code, for example, a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or the like.

The method 900 may include receiving 902 information indicating: at least one pair of channel state information reference signal resources; wherein the information comprises one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources. In certain embodiments, the method 900 includes computing 904 channel state information feedback based on the information. In one embodiment, the method 900 further comprises transmitting 906 the channel state information feedback comprising a channel state information reference signal resource indicator that indicates a pair of channel state information reference signal resources for channel measurement, wherein the channel state information feedback further comprises: one or two rank indicators; two precoder matrix indicators or two i1 indicators for the pair of channel state information reference signal resources for channel measurement; and a channel quality indicator for the pair of channel state information reference signal resources for channel measurement.

In certain embodiments, the information comprises a channel state information resource configuration. In some embodiments, the information comprises a channel state information report configuration. In various embodiments, the channel state information report configuration configures one or more pairs of channel state information reference signal resources for channel measurement.

In one embodiment, the channel state information report configuration configures one or more pairs of non-zero power channel state information reference signal resources for interference measurement, one or more channel state information interference measurement resources for inter-

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ference measurement, or a combination thereof. In certain embodiments, the channel state information report configuration configures a report quantity field comprising one or two rank indicators, two precoder matrix indicators or two i1 indicators, and a channel quality indicator.

In various embodiments, the method **900** further comprises receiving a transmission mode indicator that indicates a mode for transmitting information to a user equipment using a pair of transmission configuration indication states.

In certain embodiments, the one or two rank indicators, the two precoder matrix indicators or the two i1 indicators, and the channel quality indicator are determined based on the pair of channel state information reference signal resources for channel measurement, and a pair of non-zero power channel state information reference signal resources, a channel state information interference measurement resource, or a combination thereof. In some embodiments, the method **800** further comprises applying a quasi-co-location assumption of the pair of channel state information reference signal resources for channel measurement to a pair of non-zero power channel state information reference signal resources for interference measurement.

In various embodiments, a channel state information reference signal resource identifier of the pair of channel state information reference signal resources for channel measurement is used as the channel state information reference signal resource indicator. In one embodiment, the channel state information reference signal resource identifier occurs only once in the information.

In one embodiment, a method comprises: transmitting information indicating: at least one pair of channel state information reference signal resources; wherein the information comprises one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources.

In certain embodiments, the information comprises a channel state information resource configuration.

In some embodiments, the information comprises a channel state information report configuration.

In various embodiments, the channel state information report configuration configures one or more pairs of channel state information reference signal resources for channel measurement.

In one embodiment, the channel state information report configuration configures one or more pairs of non-zero power channel state information reference signal resources for interference measurement, one or more channel state information interference measurement resources for interference measurement, or a combination thereof.

In certain embodiments, the channel state information report configuration configures a report quantity field comprising one or two rank indicators, two precoder matrix indicators or two i1 indicators, and a channel quality indicator.

In various embodiments, the method further comprises transmitting a transmission mode indicator that indicates a mode for transmitting information to a user equipment using a pair of transmission reception points.

In one embodiment, the method further comprises receiving channel state information feedback comprising a channel state information reference signal resource indicator that indicates a pair of channel state information reference signal resources for channel measurement, wherein the channel state information feedback further comprises: one or two rank indicators; two precoder matrix indicators or two i1 indicators for the pair of channel state information reference

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signal resources for channel measurement; and a channel quality indicator for the pair of channel state information reference signal resources for channel measurement.

In certain embodiments, the method further comprises scheduling a downlink transmission based on the channel state information feedback.

In one embodiment, an apparatus comprises: a transmitter that transmits information indicating: at least one pair of channel state information reference signal resources; wherein the information comprises one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources.

In certain embodiments, the information comprises a channel state information resource configuration.

In some embodiments, the information comprises a channel state information report configuration.

In various embodiments, the channel state information report configuration configures one or more pairs of channel state information reference signal resources for channel measurement.

In one embodiment, the channel state information report configuration configures one or more pairs of non-zero power channel state information reference signal resources for interference measurement, one or more channel state information interference measurement resources for interference measurement, or a combination thereof.

In certain embodiments, the channel state information report configuration configures a report quantity field comprising one or two rank indicators, two precoder matrix indicators or two i1 indicators, and a channel quality indicator.

In various embodiments, the transmitter transmits a transmission mode indicator that indicates a mode for transmitting information to a user equipment using a pair of transmission reception points.

In one embodiment, the apparatus further comprises a receiver that receives channel state information feedback comprising a channel state information reference signal resource indicator that indicates a pair of channel state information reference signal resources for channel measurement, wherein the channel state information feedback further comprises: one or two rank indicators; two precoder matrix indicators or two i1 indicators for the pair of channel state information reference signal resources for channel measurement; and a channel quality indicator for the pair of channel state information reference signal resources for channel measurement.

In certain embodiments, the apparatus further comprises a processor that schedules a downlink transmission based on the channel state information feedback.

In one embodiment, a method comprises: receiving information indicating: at least one pair of channel state information reference signal resources; wherein the information comprises one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources.

In certain embodiments, the information comprises a channel state information resource configuration.

In some embodiments, the information comprises a channel state information report configuration.

In various embodiments, the channel state information report configuration configures one or more pairs of channel state information reference signal resources for channel measurement.

In one embodiment, the channel state information report configuration configures one or more pairs of non-zero

power channel state information reference signal resources for interference measurement, one or more channel state information reference signal resources for interference measurement, or a combination thereof.

In certain embodiments, the channel state information report configuration configures a report quantity field comprising one or two rank indicators, two precoder matrix indicators or two i1 indicators, and a channel quality indicator.

In various embodiments, the method further comprises receiving a transmission mode indicator that indicates a mode for transmitting information to a user equipment using a pair of transmission configuration indication states.

In one embodiment, the method further comprises transmitting channel state information feedback comprising a channel state information reference signal resource indicator that indicates a pair of channel state information reference signal resources for channel measurement, wherein the channel state information feedback further comprises: one or two rank indicators; two precoder matrix indicators or two i1 indicators for the pair of channel state information reference signal resources for channel measurement; and a channel quality indicator for the pair of channel state information reference signal resources for channel measurement.

In certain embodiments, the one or two rank indicators, the two precoder matrix indicators or the two i1 indicators, and the channel quality indicator are determined based on the pair of channel state information reference signal resources for channel measurement, and a pair of non-zero power channel state information reference signal resources, a channel state information interference measurement resource, or a combination thereof.

In some embodiments, the method further comprises applying a quasi-co-location assumption of the pair of channel state information reference signal resources for channel measurement to a pair of non-zero power channel state information reference signal resources for interference measurement.

In various embodiments, a channel state information reference signal resource identifier of the pair of channel state information reference signal resources for channel measurement is used as the channel state information reference signal resource indicator.

In one embodiment, the channel state information reference signal resource identifier occurs only once in the information.

In one embodiment, an apparatus comprises: a receiver that receives information indicating: at least one pair of channel state information reference signal resources; wherein the information comprises one or more identifiers, and each identifier of the one or more identifiers corresponding to the at least one pair of channel state information reference signal resources.

In certain embodiments, the information comprises a channel state information resource configuration.

In some embodiments, the information comprises a channel state information report configuration.

In various embodiments, the channel state information report configuration configures one or more pairs of channel state information reference signal resources for channel measurement.

In one embodiment, the channel state information report configuration configures one or more pairs of non-zero power channel state information reference signal resources for interference measurement, one or more channel state information interference measurement resources for interference measurement, or a combination thereof.

In certain embodiments, the channel state information report configuration configures a report quantity field comprising one or two rank indicators, two precoder matrix indicators or two i1 indicators, and a channel quality indicator.

In various embodiments, the receiver receives a transmission mode indicator that indicates a mode for transmitting information to a user equipment using a pair of transmission configuration indication states.

In one embodiment, the apparatus further comprises a transmitter that transmits channel state information feedback comprising a channel state information reference signal resource indicator that indicates a pair of channel state information reference signal resources for channel measurement, wherein the channel state information feedback further comprises: one or two rank indicators; two precoder matrix indicators or two i1 indicators for the pair of channel state information reference signal resources for channel measurement; and a channel quality indicator for the pair of channel state information reference signal resources for channel measurement.

In certain embodiments, the one or two rank indicators, the two precoder matrix indicators or the two i1 indicators, and the channel quality indicator are determined based on the pair of channel state information reference signal resources for channel measurement, and a pair of non-zero power channel state information reference signal resources, a channel state information interference measurement resource, or a combination thereof.

In some embodiments, the apparatus further comprises a processor that applies a quasi-co-location assumption of the pair of channel state information reference signal resources for channel measurement to a pair of non-zero power channel state information reference signal resources for interference measurement.

In various embodiments, a channel state information reference signal resource identifier of the pair of channel state information reference signal resources for channel measurement is used as the channel state information reference signal resource indicator.

In one embodiment, the channel state information reference signal resource identifier occurs only once in the information.

Embodiments may be practiced in other specific forms. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

The invention claimed is:

1. A method performed by a base station, the method comprising:

transmitting a channel state information (CSI) report configuration message indicating:

at least one pair of CSI reference signal (RS) resources; wherein the CSI report configuration comprises one or more identifiers and indicates two transmission configuration indicator (TCI) states, and each identifier of the one or more identifiers is mapped in the CSI report configuration to both CSI RS resources of a pair of CSI RS resources of the at least one pair of CSI RS resources.

2. A base station, comprising:

at least one memory; and

at least one processor coupled with the at least one memory and configured to cause the base station to:

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transmit a channel state information (CSI) report configuration message indicating:

at least one pair of CSI reference signal (RS) resources; wherein the CSI report configuration comprises one or more identifiers and indicates two transmission configuration indicator (TCI) states, and each identifier of the one or more identifiers is mapped in the CSI report configuration to both CSI RS resources of a pair of CSI RS resources of the at least one pair of CSI RS resources.

3. The base station of claim 2, wherein the CSI report configuration message comprises a CSI resource configuration.

4. The base station of claim 2, wherein the CSI report configuration message comprises a CSI report configuration.

5. The base station of claim 4, wherein the CSI report configuration configures one or more pairs of CSI RS resources for channel measurement.

6. The base station of claim 5, wherein the CSI report configuration configures one or more pairs of non-zero power CSI reference signal resources for interference measurement, one or more CSI interference measurement resources for interference measurement, or a combination thereof.

7. The base station of claim 5, wherein the CSI report configuration configures a report quantity field comprising one or two rank indicators, two precoder matrix indicators or two i1 indicators, and a channel quality indicator.

8. The base station of claim 5, wherein the at least one processor is configured to cause the base station to transmit a transmission mode indicator that indicates a mode for transmitting information to a user equipment using a pair of transmission reception points.

9. The base station of claim 2, wherein the at least one processor is configured to cause the base station to receive CSI feedback comprising a CSI reference signal resource indicator that indicates a pair of CSI reference signal resources for channel measurement, wherein the CSI feedback further comprises:

one or two rank indicators;

two precoder matrix indicators or two i1 indicators for the pair of CSI reference signal resources for channel measurement; and

a channel quality indicator for the pair of CSI reference signal resources for channel measurement.

10. The base station of claim 9, wherein the at least one processor is configured to cause the base station to schedule a downlink transmission based on the CSI feedback.

11. A user equipment (UE), comprising:

at least one memory; and

at least one processor coupled with the at least one memory and configured to cause the UE to:

receive a channel state information (CSI) report configuration message indicating:

at least one pair of CSI reference signal (RS) resources;

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wherein the CSI report configuration comprises one or more identifiers and indicates two transmission configuration indicator (TCI) states, and each identifier of the one or more identifiers is mapped in the CSI report configuration to both CSI RS resources of a pair of CSI RS resources of the at least one pair of CSI RS resources.

12. The UE of claim 11, wherein the CSI report configuration message configures one or more pairs of CSI reference signal resources for channel measurement.

13. The UE of claim 12, wherein the CSI report configuration message configures one or more pairs of non-zero power CSI reference signal resources for interference measurement, one or more CSI interference measurement resources for interference measurement, or a combination thereof.

14. The UE of claim 12, wherein the CSI report configuration message configures a report quantity field comprising one or two rank indicators, two precoder matrix indicators or two i1 indicators, and a channel quality indicator.

15. The UE of claim 12, wherein the at least one processor is configured to cause the UE to receive a transmission mode indicator that indicates a mode for transmitting information to a user equipment using a pair of transmission configuration indication states.

16. The UE of claim 11, wherein the at least one processor is configured to cause the UE to transmit CSI feedback comprising a CSI reference signal resource indicator that indicates a pair of CSI reference signal resources for channel measurement, wherein the CSI feedback further comprises: one or two rank indicators;

two precoder matrix indicators or two i1 indicators for the pair of CSI reference signal resources for channel measurement; and

a channel quality indicator for the pair of CSI reference signal resources for channel measurement.

17. The UE of claim 16, wherein the one or two rank indicators, the two precoder matrix indicators or the two i1 indicators, and the channel quality indicator are determined based on the pair of CSI reference signal resources for channel measurement, and a pair of non-zero power CSI reference signal resources, a CSI interference measurement resource, or a combination thereof.

18. The UE of claim 16, wherein the at least one processor is configured to cause the UE to apply a quasi-co-location assumption of the pair of CSI reference signal resources for channel measurement to a pair of non-zero power CSI reference signal resources for interference measurement.

19. The UE of claim 16, wherein a CSI reference signal resource identifier of the pair of CSI reference signal resources for channel measurement is used as the CSI reference signal resource indicator.

20. The UE of claim 19, wherein the CSI reference signal resource identifier occurs only once in the information.

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